

Fog, Friction, Delay and the Failure of Bounded Rationality Embodied Cognition: A formal study of generalized psychopathology

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Abstract

A pared-down, best-case, mathematical model of the evolutionarily ubiquitous cognition/regulation dyad finds that its failure on ‘Clausewitz landscapes’ of fog, friction, and adversarial intent is – inevitably – likewise ubiquitous. From biological subsystems to individuals, institutions, intelligent machines, and their many composites, the expression of culture-bound ‘psychopathology’ under stress and uncertainty is not a bug. It is an inherent feature of real-world cognition across all possible modes, scales, and levels of organization, a feature for which prevention and remediation remain dire challenges. The probability models developed here, founded on the asymptotic limit theorems of information and control theories, allow construction of new statistical tools for the analysis of experimental and observational data regarding such cognitive failure and its possible address.

Keywords: artificial intelligence, cognitive failure, control theory, culture-bound syndrome, information theory, noise, phase transition, selection pressure, stress

Tactics without strategy is the sound before defeat.

— Sun Tzu

Culture eats strategy for breakfast.

— Peter Drucker

Culture is everything.

— Lou Gerstner

Culture is as much a part of human biology as the enamel on our teeth.

— Robert Boyd

‘elementary’ immune system is cognitive, exercising choice-of-action in response to internal or external signals, choice that formally reduces uncertainty. The reduction of uncertainty, in the context of the asymptotic limit theorems of information and control theories, implies that a cognitive process defines a ‘dual information source’ instantiating that reduction. Much of what we will do here follows directly from this ‘simple’ insight. We will attempt to hold closely to those theorems, much in the sense that ‘statistical physics’ remains bounded by – and can indeed be expressed in terms of – the Central Limit, Ergodic, and related theorems (Khinchin 1949).

Inherent to what we will do is the growing consensus across cognitive science, neuroscience and information theory that cognitive stability requires an intimate pairing of cognitive process with a parallel regulatory process:

- Immune systems must limit their attack on ‘self’ to routine maintenance: T-cells are paired with T-regulatory cells. More generally, as Grossman et al. (2023) put it,

The regulation of immunity and homeostasis is concomitant, flexible, and smart... [T]o maintain the functional integrity of tissues, the body combines the information processing capacities of the brain with those that are possessed by the immune system. This requires that we selectively and cautiously apply to immunity cognitive science concepts and tools.

- Similarly, blood pressure must be kept within tolerable limits, even in the face of extreme effort.
- Institutional cognition is bounded by internal doctrine and the constraints of law and embedding culture.

1. Introduction

Cognitive phenomena of every nature, at every scale, at every level of organization, have emerged from, are embedded and participate in, are bounded by, evolutionary process. While, as Dobzhansky put it, ‘Nothing in biology makes sense except in the light of evolution’, selection pressures also sculpt social, cultural, and institutional phenomena in the usual manner of path-dependent historical trajectories. Elaborate, emerging, composite systems involving machine cognition are not spared.

Maturana and Varela (1980) go so far as to hold that the living state itself is cognitive at every scale and level of biological organization. Atlan and Cohen (1998) assert that even the

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- Theories of embodied cognition and predictive coding, in address of schizophrenia and hallucination, explore how the interplay of top-down cognitive/regulatory and bottom up sensory/environmental processes is inherently unstable (e.g., Fan 2024; Gao et al. 2022; Tschacher et al. 2027; Sayood 2018).

The necessary pairing of embodied cognition with regulation for evolutionarily-determined function will be a central focus of what follows, in the context of the inherent instability that afflicts all such systems. That is, embodied cognition, defined by interaction with an embedding environment having its own distinct and characteristic ‘native language’ producing what the theorists call ‘topological information’ at its own rate, is rendered inherently unstable by that very interaction.

Thus an agent of embodied cognition must be paired with a regulatory agent that provides control information at a rate greater than the environmental roadway is generating its impinging topological information. One standard picture is of a vehicle driven at high speed on a rough surface. The driver must brake, shift, and steer faster than the surface imposes a topology of bumps, twists, and potholes, given the chosen speed. Formal study is via the Data Rate Theorem of control theory (Nair et al. 2007) whose constraints, we show, can be directly implemented in formal models of embodied cognition.

We begin with a brief restatement of contemporary ‘Bounded Rationality’ perspectives in terms of Rate Distortion Control Theory, under the constraint of the Data Rate Theorem, providing a much stripped-down version of more elaborate theory that incorporates such terms as ‘groupoid symmetry-breaking phase transitions’ (Wallace 2022, 2023, 2025a, b, 2026). This reformulation allows a relatively intuitive extension of theory to the burdens of Clausewitzian ‘fog and friction,’ given different imperatives for system stabilization.

Most particular in this regard is a contrast between stabilization of underlying structure vs. stabilization of immediate perception. This dichotomy should be familiar to those who have been involved with the coupled pathologies of individual and institutional behavior under selection pressures. One is, perhaps, reminded of the term ‘Darwin Award’.

Again, current and emerging composite systems deeply enmeshed with machine cognition cannot, and will not, be spared. For these, embodiment, indeed, is a best-case scenario. Without it, ‘artificial intelligence’ can, ultimately, only express bizarre and hallucinatory dreams of reason.

2. An outline of method

Resources

The systems we study are embedded in-and-of an environment that includes themselves. There must be at least three resource streams necessary for their correct and correctable function:

- The rate at which elements of the system can communicate with each other, characterized by a channel capacity C .

- The rate at which sensory/intelligence information is available from the embedding milieu, characterized by a channel capacity $\mathcal{H}$.

- The rate at which material/material usable ‘free energy’ resources are provided, a rate $\mathcal{M}$. For an organism, this might be measured by the rate at which metabolic free energy is available.

These – and possibly other – rates cross-correlate and interact, generating a (minimally) 3×3 matrix $\mathbf{Z}$. An $n \times n$ matrix will have n scalar measures r_i invariant under similarity transformations that are determined by the standard expression

$$p(\gamma) = \det[\mathbf{Z} - \gamma \mathbf{I}] = \sum_{k=0}^n (-1)^k r_k \gamma^{n-k} \quad (1)$$

$\mathbf{I}$ is the identity matrix, ‘det’ the determinant, and γ a real-valued parameter. The first is the matrix trace, and the last the determinant. We construct a scalar resource rate measure from the r_k .

An elementary such index might simply be $Z = C \times \mathcal{H} \times \mathcal{M}$, but matters are not likely quite that simple.

We assume a first-order model, allowing that a single scalar index $Z(r_1, \dots, r_n)$ can be constructed that sufficiently represents information and material resource rates. This argument is analogous to principal component analysis based on a correlation matrix.

Rate Distortion Control Theory

An ‘embodied’ formalism emerges via a ‘Rate Distortion Control Theory’ based on the Rate Distortion Theorem (RDT) (Cover and Thomas 2006; Khinchin 1957). The RDT addresses transmission under conditions of noise over an information channel. There will be, for that channel and a particular scalar measure of average distortion D between what is sent and what is received, a minimum necessary channel capacity $R(D)$.

The theorem asks, in a sense, what is the ‘best’ channel for transmission of a message with the least possible average distortion. $R(D)$ can be defined for nonergodic information sources via a limit argument based on the ergodic decomposition of a nonergodic source into a ‘sum’ of ergodic sources. The RDT can be reconfigured in a control theory context, if we envision a system’s topological information from the Data Rate Theorem as another form of noise, adding to the average distortion D . See figure 1.

The punctuation implied by the Data Rate Theorem (DRT) (Nair et al. 2007) happens if there is a critical maximum average distortion beyond which control fails.

Embodied cognition, characterized by continuous interaction with a real world having its own intents and capabilities – as indexed by the distortion between what is wanted and what is observed – is inherently a phenomenon of ‘bounded rationality’ in the sense of Ortega and Braun (2013). Cognition without such embodied interaction, however, as Dreyfus (1972), Fjelland (2020), and many others have noted, invokes a narrow boundedness without rationality constraints.

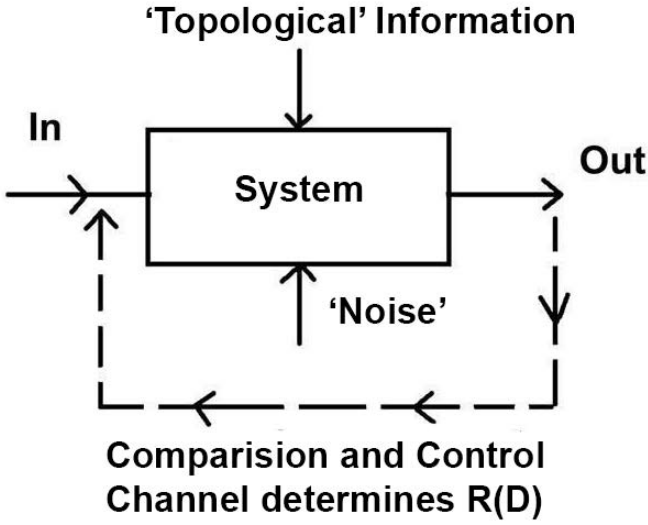


Figure 1: The Rate Distortion Control Theory model. An inherently unstable system's 'topological information', according to the Data Rate Theorem (Nair et al. 2007), is added to the effects of 'noise' to determine the minimum channel capacity $R(D)$ needed to assure transmission with average distortion D . The punctuation implied by the DRT manifests if there is a critical maximum average distortion beyond which control simply fails.

The reader should be aware that what we do here, while invoking the asymptotic limit theorems of information and control theories, does so in a very limited manner. At best there appear, almost by magic, analogs to Fisher Zero phase transitions (Dolan et al. 2001) driven by the occurrence of imaginary-valued terms in what should be real-valued quantities, in particular temperature analogs. There is far more hidden here, i.e., the 'groupoid symmetry-breaking phase transitions' that underlie fundamental matters of cognition at every scale and level of organization (e.g., Wallace 2022, 2023). The term is common in string theory, and this should be a warning: Here there be Tygers. We have, for pedagogic purposes, pruned the mathematics to most directly characterize the basic conundrum. The engineering design and implementation of corrective measures will be quite another matter, requiring tools at the very outer boundaries of current formal theory. Yes, cognitive science really is even more challenging than cutting-edge physics, Tech Bro venture capital pitches to the contrary.

3. A 'simple' model

Consider a binary, paired cognition/regulation system governed by some fundamental, characteristic probability distribution ρ acting across the two structures – the 'cognitive' and the 'regulatory'. These are characterized by rate distortion channel capacities R_j , $j = 1, 2$ at 'temperatures' $g_j(Z_j)$, so that, following longstanding conventional argument (e.g., Landau and Lifshitz 2007) the probability of 'state' j is determined as

$$P_j = \frac{\rho(R_j/g_j(Z_j))}{\rho(R_1/g_1(Z_1)) + \rho(R_2/g_2(Z_2))} \quad (2)$$

where the precise form of the 'temperature' functions $g_j(Z_j)$ will, somewhat surprisingly, remain undetermined.

At this point, we do not pursue higher-order 'ideotypic network' back-and-forth iterations of the underlying approach.

An essential subtheme of this work is that 'information' is not an entropy, in spite of Shannon uncertainty taking an algebraic expression as similar to a simple entropy for a stationary, ergodic system. Rather, following Feynman (2000) and Bennett (1982), information is, in fact, best represented as a form of free energy. Feynman (2000) in particular explores Bennett's imaginary machine that converts the information within a transmitted message into useful work, free energy. Here, we iterate that argument, recognizing the denominator of Eq.(2) as a *partition function* (Landau and Lifshitz 2007; Feynman 1998), leading to an iterated free energy F

$$\rho(F/G) = \rho(R_1/g_1(Z_1)) + \rho(R_2/g_2(Z_2)) \quad (3)$$

where, again, the g_j are local modular temperature analogs and G is an 'effective system temperature' that must – and surprisingly, can – be determined from first principles (Wallace 2026).

Here, we take G as the geometric average

$$G \equiv \sqrt{g_1(Z_1)g_2(Z_2)} \quad (4)$$

The arithmetic average leads, ultimately, to much the same result.

As an opener, we assume the first-order model $g(Z) = g \times Z$, where the constant g is an affordance index that determines how well the resource rate Z can actually be used, and, for convenience, take $\rho(x)$ as the Boltzmann Distribution, solving for F as

$$F = - \left(\sqrt{g_1 Z_1 g_2 Z_2} \right) \ln \left(e^{-\frac{R_1}{g_1 Z_1}} + e^{-\frac{R_2}{g_2 Z_2}} \right) \quad (5)$$

Dynamics are defined in second order at the nonequilibrium steady state according to an Onsager nonequilibrium thermodynamics model (de Groot and Mazur 1984),

$$S \equiv -F + Z_1 dF/dZ_1 + Z_2 dF/dZ_2 \\ dZ_j/dt \propto \partial S/\partial Z_j = 0, \quad j = 1, 2 \quad (6)$$

where S , as the Legendre Transform of F (Pettini 2007), is an entropy-analog, and we omit the 'diffusion coefficient' in the second line of Eq.(6).

Solving the two relations $\partial S/\partial Z_j = 0$ for the R_j leads to three solution sets for the nonequilibrium steady state (nss) – i.e., homeostatic/allostatic – conditions. The physically non-trivial one is the equipartition relation

$$\frac{R_1}{g_1 Z_1} = \frac{R_2}{g_2 Z_2} \equiv X \quad (7)$$

We view this result as a canonical *iterated equipartition* assumption that extends the 'equipartition' of the Shannon McMillan Theorem (SMT) to certain nonergodic information and control systems (Cover and Thomas 2006; Khinchin 1957). Following Khinchin (1957), for nonergodic information sources,

each individual emitted ‘path’ eventually converges on its own particular value of an uncertainty measure splitting criterion *not given as a Shannon ‘entropy’*. Only for ergodic stationary sources does the uncertainty measure have the same mathematical expression as the familiar ‘entropy’. See Wallace (2022) for the ‘nonergodic’ details. Here, via Eq.(7), we have recovered an iterated version of the SMT that we will further generalize.

Also implicit is a partitioning of system dynamics according to a ‘Born-Oppenheimer’ approach in which ‘fast’ and ‘slow’ modes are initially taken as effectively decoupled. The analog is the assumption of slow nuclear oscillations in a molecule, compared with the much faster electronic processes that are assumed independent of, and not affecting, those oscillations. Higher order interactions can then be added as ‘perturbations’ of the decoupled model, something we do not pursue here. The methodology has long been standard.

We extend the argument for equipartition in cognitive systems under nonequilibrium steady state conditions, making an analog to the equilibration between excitation modes in a complex molecular, or indeed, macroscopic, structure affected by powerful, random excitations. Then Eq.(3), defining the ‘temperature’ G , becomes

$$\rho\left(\frac{F(X)}{G(X)}\right) = \rho(X) + \rho(X) = 2\rho(X)$$

$$G(X) = \frac{F(X)}{\rho^{-1}(2\rho(X))} \quad (8)$$

where the factor of ‘2’ may become a general parameter under complicated modes of equipartition (e.g., highly reflexive ideotypic networks).

A first-cut model of system dynamics can be extracted via a simplified form of the Onsager Approximation introduced above. Again define an entropy-analog as the Legendre Transform of the iterated free energy F , and take $dX/dt \propto dS/dX$, leading to

$$S \equiv -F + XdF/dX$$

$$dX/dt \propto dS/dX = Xd^2F(X)/dX^2 \quad (9)$$

where, again, the ‘diffusion coefficient’ has been set equal to one, remembering that information systems, unlike physical systems, are not microreversible: ‘eht’ does not have the same probability as ‘the’ in English. Hence, for multidimensional systems, there are no ‘Onsager reciprocal relations’.

To determine G from Eq.(8), we must fix $F(X)$.

We introduce further stochastic burden, beyond that driving ‘equipartition’, using the stochastic differential equation

$$dX_t = (X_t d^2F/dX^2)dt + \sigma h(X_t)dB_t \quad (10)$$

The second term represents ‘volatility’ in the context of Brownian white noise dB_t .

The Gordian Knot of inference can be cloven by asking what, in particular, is actually being ‘regulated’.

Two possibilities seem more-or-less obvious.

The first is regulation of the foundation structure itself, as represented by the basic underlying probability distribution $\rho(X)$.

The second is regulation of the perceived system condition as determined by some Stevens Law sensation analog, $Q(X) = k(X)^a$, $0.5 \leq a \leq 3.5$.

We can ‘easily’ find appropriate forms of $F(X)$ by applying the Ito Chain Rule (Cyganowski et al. 2002) to determine the nonequilibrium steady state conditions $\langle d\rho_t \rangle = 0$ and $\langle dQ_t \rangle = 0$.

For any function $V(X)$, given the expression for volatility as $h(X)$ of Eq.(10), demanding that $\langle dV_t \rangle = 0$ under the Ito Chain Rule – the nonequilibrium steady state – defines $F(X)$ as

$$F(X) = \int \int -\frac{\sigma^2 h(X)^2 \left(\frac{d^2}{dX^2} V(X)\right)}{2X \left(\frac{d}{dX} V(X)\right)} dXdX + C_1 X + C_2 \quad (11)$$

The calculation is elementary but tedious and best done using a computer algebra program (e.g., Cyganowski et al. 2002).

Here, we take $h(X) = X$, defining ‘ordinary’ volatility.

The results for the Boltzmann Distribution $\rho(x) = \exp[-x]$ and Q as the Stevens Law are, respectively,

$$F = \frac{1}{12}\sigma^2 X^3 + C_1 X + C_2$$

$$F = \frac{1}{4}\sigma^2(1-a)X^2 + C_1 X + C_2 \quad (12)$$

with, for the Boltzmann Distribution,

$$G(X) = -\frac{F(X)}{\ln(2) - X} \quad (13)$$

For the nonmonotonic second-order Erlang Distribution representing an iterated, two-step, Boltzmann model, then $\rho(x) = x \exp[-x]$, and calculation finds, as might be expected, two solutions for G :

$$G(X) = -\frac{F(X)}{W\left(m, -\frac{2X}{e^x}\right)} \quad (14)$$

where $W(m, x)$ is the transcendental Lambert W-function of orders $m = 0, -1$ that satisfies the relation

$$W(m, x) \exp[W(m, x)] = x$$

For $m = 0$, it is real-valued if $x \geq -\exp[-1]$, and for the $m = -1$ branch, only if $-\exp[-1] \leq x \leq 0$, expressing the two aspects of the nonmonotonic distribution, i.e., constrained to the left of the peak, unconstrained to the right.

And so on. That is, the approach becomes necessarily complicated under the endless branching of real-world structural and functional diversity, a matter that can be pruned only by empirical observation and experimentation.

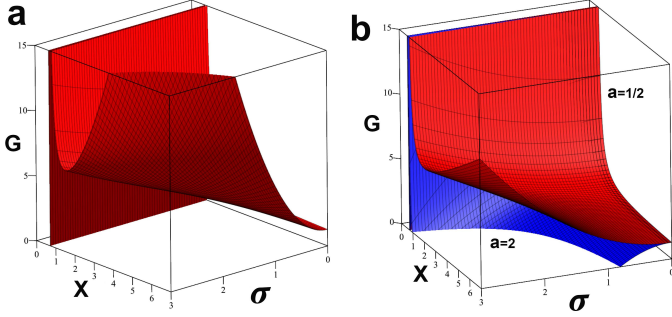


Figure 2: Calculation of the temperature analog G under a Boltzmann Distribution, setting $C_1 = C_2 = 1$ for all. **a.** The $\langle d\rho_t \rangle = 0$ nonequilibrium steady state. The system is strongly ‘channeled’ according to X and σ . **b.** The $\langle dQ_t \rangle = 0$ nss under Stevens Law, taking $a = 1/2, 2$. The $a = 2$ mode displays highly punctuated phase transitions to inherent instability.

4. Patterns and predicaments

In figure 2, we restrict calculation of the temperature analog G to the Boltzmann Distribution, looking at $\langle d\rho_t \rangle = 0$ in figure 2a, and to Stevens Law examples under the Boltzmann Distribution in figure 2b, setting $\langle dQ_t \rangle = 0$ for $a = 1/2, 2$, where the term in σ^2 changes sign. For all, $C_1 = C_2 = 1$.

Both cases show ‘hallucination’ at low values of X .

In figure 2a, the Boltzmann nss solution channels the system into a relatively narrow valley of accessibility that rises with increasing ‘noise’ σ . The ‘wings’ are likely unsustainable.

For figure 2b, the system bifurcates according to the exponent a , first following an analog of figure 2a, but, as the exponent increases, switching to a highly punctuated ‘phase transition’ mode of inherent instability.

Equations 8 and 11 permit more general study. For example, an extension of Eq.(8) might, in fact, involve different probability distributions: Consider, for example, a system that must navigate a multi-stage exponential-distribution decision tree, typically represented in terms of Erlang Distributions

$$\rho(m, k, x) = \frac{m^k x^{k-1} \exp[-mx]}{(k-1)!} \quad (15)$$

$$\langle x \rangle = \frac{k}{m}$$

where k represents the tree depth.

An n -layer decision tree then gives

$$\exp[-F/G] = \sum_{k=1}^n \rho(1, k, X) = \frac{\Gamma(n, X)}{\Gamma(n)} \rightarrow 1 \quad (16)$$

where Γ is the Gamma function, taking F as the two expressions of Eq.(12). Note that the limit $n \rightarrow \infty$ is highly pathological, requiring an ‘infinite’ temperature.

In contrast, for ideotypic back-and-forth action/control networks of order n , taking $\rho(k, X)$ as the Exponential, Lomax, and Levy Distributions, then

$$\begin{aligned} \exp[-F/G] &= \sum_{k=1}^n k \exp[-kX] = \\ &= \frac{((-1-n)e^X + n)e^{-Xn} + e^X}{(e^X - 1)^2} \rightarrow \frac{1}{4 \sinh(X/2)^2} \\ \exp[-F/G] &= \sum_{k=1}^n \frac{k}{(1+X)^{k+1}} = \\ &= \frac{1 + (-1 + (-1-n)X) \left(\frac{1}{X+1}\right)^{n+1}}{X^2} \rightarrow \frac{1}{X^2} \\ \exp[-F/G] &= \sum_{k=1}^n \frac{\sqrt{\frac{k}{2\pi}} \exp[-k/(2X)]}{X^{3/2}} \rightarrow \\ &= \frac{\text{Li}_{-1/2}(e^{-\frac{1}{2X}}) \sqrt{2}}{2 \sqrt{\pi} X^{3/2}} \end{aligned} \quad (17)$$

where $\text{Li}_{-1/2}$ is the Polylog function of order $-1/2$. The expressions are possibly multiplied, on the right hand side, by an appropriate scale parameter. Note that the Levy Distribution is ‘fat-tailed’ and does not have finite first moments.

Both the serial Erlang of finite order, and first two Ideotypic approaches, lead to distorted versions of figure 2 under the same boundary values on the two forms of F . The Levy Distribution pattern is left as an exercise.

Other compound models are, of course, possible.

Such explorations, as with any and all probability models and statistical tools, must ultimately be pruned and made useful by continuous feedback with empirical observation and experiment.

5. Clausewitz’ revenge: Fog, friction and delay

A standard first approximation introducing both friction and delay is to assume that X is strongly constrained as

$$dX(t)/dt = \beta - \alpha X(t - \tau) \rightarrow \beta/\alpha \quad (18)$$

where α represents friction and τ a fixed delay.

After some algebra,

$$\begin{aligned} X(t) &= \frac{\beta}{\alpha} (1 - \exp[S t]) \\ S &\equiv \frac{W(m, -\alpha\tau)}{\tau} \\ dX/dt &= S \left(X - \frac{\beta}{\alpha} \right) \\ dX_t &= S \left(X_t - \frac{\beta}{\alpha} \right) dt + \sigma X_t dB_t \end{aligned} \quad (19)$$

for $m = 0, -1$ in the Lambert W-function, providing a necessary condition for stability as

$$\alpha\tau < \exp[-1] \quad (20)$$

with S always negative when real-valued.

Carrying out the Ito Chain Rule stabilizations on $\rho(X) = \exp[-X]$ and $Q(X) = kX^a$ gives even more convoluted stability conditions arising from the nss relations

$$\hat{X} = \frac{S\alpha \pm \sqrt{S\alpha(-2\beta\sigma^2 + S\alpha)}}{\alpha\sigma^2}$$

$$\hat{X} = \frac{S\beta/\alpha}{\frac{\sigma^2(a-1)}{2} + S} \quad (21)$$

remembering again that S is negative when real-valued.

Details are left as an exercise.

Clearly, imposing friction and delay on the approach to nonequilibrium steady state equipartition does not enhance system stability.

‘Simple’ distributed delay can be addressed using the Laplace Transform to solve

$$dX/dt = \beta - \alpha \int_0^t \rho(\tau)X(t - \tau)d\tau \quad (22)$$

for $X(t)$, where $\rho(\tau)$ is the distribution of the delay effect. Explicit solutions tend to show considerable oscillation as $X(t) \rightarrow \beta/\alpha$.

Deriving the analog of Eq.(12) to find the effect of ‘noise’ is, however, not so straightforward, given that the general treatment of stochastic differential delay equations remains on the cutting edge of applied mathematics (e.g., Mao 1996; Guiloulic et al. 1999; Babasola et al. 2023).

A specialized approach for the exponential distribution $\rho(x) = m \exp[-mx]$, with $\langle x \rangle = 1/m$, is to reprise the third part of Eq.(19), again applying the Laplace Transform method:

$$dX/dt = \beta - \alpha \int_0^t m \exp[-m\tau]X(t - \tau)d\tau$$

$$\equiv S \left(X(t) - \frac{\beta}{\alpha} \right)$$

$$X = \beta \left(\frac{e^{-\frac{m(S+\alpha)t}}{S}} + \frac{1}{\alpha} \right)$$

$$S = -\frac{m}{2} \pm \frac{\sqrt{m^2 - 4\alpha m}}{2} \quad (23)$$

Since $\langle \tau \rangle = 1/m$, a necessary condition for real-valued S is

$$\alpha < \tau > \leq \frac{1}{4} \quad (24)$$

analogous to Eq.(20).

The results of Eq.(21) can then be imposed.

Recalling Eq.(15), we next consider the $k = 2$ Erlang Distribution, characterizing a sequence of k exponentially distributed decision processes.

A similar line of argument leads to the stability condition

$$\alpha < \tau > \leq \left(\frac{2}{3} \right)^3 \approx 0.2963 \quad (25)$$

For $k = 3$,

$$\alpha < \tau > \leq \left(\frac{3}{4} \right)^4 \approx 0.3164 \quad (26)$$

Argument by induction for the Erlang Distribution system of order k rears itself as the conjecture

$$\alpha < \tau > \leq \left(\frac{k}{k+1} \right)^{k+1} \rightarrow \exp[-1] \quad (27)$$

reminiscent of Eqs.(20) and (24), since the Exponential Distribution is simply the Erlang of order one.

The Mathematical Appendix attempts to reify this result with an approximation for a general $\rho(x)$ that uses Jensen’s Inequality.

The method of Eq.(22) et seq. is, however, not necessarily general, and the shoals, eddies and riptides uncovered by Mao (1996) and others – essentially the full-bore fog and friction of Clausewitz – will continue to challenge and confront the intrepid explorer of embodied cognition and its discontents.

Even Tech Bro billionaires will not be spared.

6. The dynamics of constraint

Some heuristics.

Suppose the system of interest is composite, having n cognition-regulation pairs, each associated with a nonequilibrium steady state index rate X_i as above. Suppose there are limits on both the X_i and the times T_i under which the individual component pair can react. That is, both resources and action times are limited. We define a Lagrangian based on some cognition-regulation pair fitness measure L_i , for example cognition or reaction rate. The Lagrangian becomes

$$\mathcal{L} = \sum_i L_i + \lambda \left(X - \sum_i X_i \right) + \mu \left(T - \sum_i T_i \right)$$

$$\partial \mathcal{L} / \partial Z = \lambda, \quad \partial \mathcal{L} / \partial T = \mu \quad (28)$$

Again heuristically, $dZ/dT \sim \mu/\lambda$. λ and μ are economic shadow prices in the sense of Jin et al. (2008) and Robinson (1993) that index environmental burden, in a large sense.

Supposing, as above, that $dX/dT = f(X)$, then, again crudely,

$$dZ/dT = \frac{\mu}{\lambda} \equiv \omega = f(X)$$

$$X(T) = f^{-1}(\omega) \quad (29)$$

For example, assuming again $f(X) = \beta - \alpha X$ or $S(X - \beta/\alpha)$, with S defined from Eq.(19), then

$$X = \frac{\beta - \omega}{\alpha}$$

$$X = \frac{\beta}{\alpha} + \frac{\omega}{S} \quad (30)$$

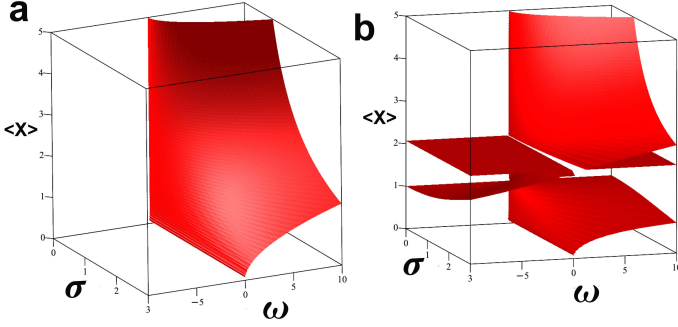


Figure 3: The synergism of noise and environmental constraint: **a.** Boltzmann Distribution representing ‘mission command’ doctrine. **b.** Second-order Erlang Distribution, representing a two-step ‘detailed command’ doctrine.

remembering that $S \leq 0$ and $X_i \leq X$.

As the environmental burden, the shadow price ratio ω rises, the system eventually collapses, whatever the chosen fitness measure.

We can, in some contrast to these results, extend the argument to a ‘noisy’ system – incorporating something of Clausewitz’ revenge – by imposing the SDE

$$dX_t = \omega dt + \sigma X_t dB_t \quad (31)$$

to determine the nonequilibrium steady state stability conditions for the Boltzmann Distribution, Stevens Law, and general expressions (i.e., $\langle dQ_t \rangle = 0$, etc.), again under the Ito Chain Rule, as in Eq.(21). Then

$$\begin{aligned} \hat{X} &= \frac{\sqrt{2}\omega}{\sigma} \\ \hat{X} &= \frac{\omega}{\frac{\sigma^2}{2}(1-a)} \\ \frac{\sigma^2}{2\omega} &= -\frac{\frac{d}{dX}Q(X)}{X^2\left(\frac{d^2}{dX^2}Q(X)\right)} \end{aligned} \quad (32)$$

where the latter expression is to be back-solved for $\hat{X}$.

In figure 3, from this perspective, we compare the Boltzmann and Second Order Erlang Distributions, respectively $\rho(x) = \exp[-x]$, $x \exp[-x]$. These can be taken as representing one-step ‘Mission Command’ and two-step ‘Detailed Command’ dynamics under the synergism of noise σ with environmental constraint ω . The message seems obvious.

In these examples, *iterated* ratios of the environmental shadow price ratio to noise dominate dynamics, rather than the shadow price ratio alone. Matters have become even more confounding.

The Appendix to Wallace (2026, Ch.1) permits expansion of an analogous but still more convoluted model.

These are deep and turbulent waters that, realistically, can only become more difficult to navigate.

7. The ‘cultural artifact’ perspective

The equipartition model of Eqs.(2)-(7) is not confined to a Rate Distortion Control Theory treatment of bounded rational-

ity. Rather, it permits the ‘breaking out’ of some particularly powerful effector from an ensemble of what is influenced by it.

To reiterate, cognition implies choice, choice reduces uncertainty, and the reduction of uncertainty implies the existence of an information source ‘dual’ to the cognitive process, a de-facto ‘language’ that, in fact, instantiates it (Atlan and Cohen 1998; Wallace 2025b, Ch.4). Any human-associated cognitive system – from individual human mind to ‘AGI’ to empire – is an artifact under the aegis of some particular culture.

There is, then, a natural bifurcation.

First, culture can be envisioned as an information source, say Y_C , having its own grammar and syntax under a source uncertainty $H(Y_C)$.

Second, any evolutionarily-sustained cognitive artifact within that culture can then be associated with a joint information source embracing

- a set $\{Y_i, Y^i\}$ of linked cognitive and regulatory submodules within the artifact
- an information source Y_V representing the grammar and syntax of the season-like embedding environmental or social context
- an information source Y_Δ representing the regularity of ‘large deviations’, in the sense of Dembo and Zeitouni (1998) and Champagnat et al. (2006).

The resulting convolution can be expressed in terms of a joint source uncertainty $H(\{Y_i, Y^i\}, Y_V, Y_\Delta)$.

Eqs.(2)-(7) can then be reexpressed in terms of these two information sources, with the two forms of ‘ H ’ replacing the ‘ R ’. The rest more or less follows.

Indeed, following Wallace (2025b, Sec. 4.6) it is possible to define a distortion measure across such systems in terms of the different H -measures themselves, leading, full circle, to an iterated Rate Distortion Theorem treatment.

A central implication of this analysis is that the generalized psychopathologies afflicting cognitive cultural artifacts – from individual minds and AI entities to the social structures and formal institutions that incorporate them – are all effectively culture-bound syndromes (Wallace 2015, 2018).

One can, however, by stepping further out – at the expense of an iterated ‘simplification’ – rewrite cognitive dynamics in terms of the ‘full-system joint source uncertainty’

$$H_S \equiv H_S(\{Y_i, Y^i\}, Y_V, Y_\Delta, Y_C)$$

This information source is, in general, nonergodic, i.e., path dependent, so that each high probability grammatical and syntactical behavioral sequence $x = \{x_1, x_2, \dots\}$, where the x_j are taken from an available ‘tactical alphabet’, converges on its own individual source uncertainty value $H_S(x)$ that cannot be represented as a ‘Shannon entropy’ across a fixed probability set (Khinchin 1957).

The full set of possible high probability paths x , it is assumed, can be aggregated into distinct equivalence classes according to the ‘game’ being played. Equivalence classes, in turn, define a *groupoid* by standard arguments (Weinstein 1996). A groupoid is a generalization of the idea of a symmetry group,

in which there are multiple possible ‘unit/identities’, but no necessary product between all possible groupoid elements.

The simplest such is the disjoint union of a set of ‘ordinary’ symmetry groups. Think of a bag of ball bearings. Putting a ding in a single bearing causes a full-scale shift in the groupoid symmetry of the entire bag, a groupoid symmetry-breaking phase transition indexing the physical ‘ding’.

One then becomes generally interested in ‘groupoid symmetry-breaking phase transitions’ that define shifts between ‘games’ being played, or, within the same game, different strategies pursued. Or the sudden expression of different forms of behavioral pathology under stress or resource constraint. Details of the perspective can be found in Wallace (2020, 2022, 2023).

8. Discussion

As the theoretical ecologist E.C. Pielou (1977) argues, mathematical models of complex social, biological, and other such phenomena best serve to raise questions for empirical study in a process of continuing dialog with such study. A central question raised by what we have done here regards the conditions for stability of the ubiquitous – and necessary – cognition/regulation dyad for systems instantiating bounded rationality embodied cognition on what might be called Clausewitz landscapes of fog, friction, and skilled adversarial intent.

We first focused on ‘fog’, expressed by the parameter σ and the function $h(X)$ in Eq.(10). The synergistic effects of friction and delay were pursued by standard methods in Section 5, compounding Clausewitz’ revenge.

Adversarial intent can be expressed as both strategic increase of the burdens of fog and friction and as exercises in changing the rules of the underlying game, e.g., Japanese use of machine guns and barbed wire in the 1905 war with Russia, use of drones in the Armenia-Azerbaijan Conflict, countering of US B-52 carpet bombing by Vietnamese and Afghan cultural persistence, and so on.

We have also explored a fundamental disconnect in dynamics engendered by divergent foci on stabilizing underlying structure, represented as $\langle d\rho_t \rangle = 0$, vs. stabilizing ‘perception’ $\langle dQ_t \rangle = 0$, where Q is something like the Stevens Law. The political term ‘kicking the can down the road’ comes to mind, as perhaps do other, far less polite, euphemisms.

The pattern of often highly punctuated pathologies encountered in this study is not comforting for those tasked with the maintenance and remediation of individual and institutional function: medicine, public health, and governance. Exploration of other probability distributions and/or perception models only leads deeper into the morass.

Any AI entity, up to and including an ‘artificial general intelligence’, will be encumbered by that same morass: we have carried out a very general best-case analysis of a highly regulated, highly optimized system.

In sum, failure of bounded rationality embodied cognition under stress – the expression of de-facto culture-bound psychopathology – is not a bug. It is an inherent feature of the evolutionarily-sculpted cognition/regulation dyad.

Welcome to our world.

9. Mathematical Appendix: Jensen’s Inequality

The assumption is that, for a probability distribution $\rho(\tau)$ on $[0, \infty)$ having a finite mean $\langle \tau \rangle$,

$$dX/dt = \beta - \alpha \int_0^t \rho(\tau)X(t-\tau)d\tau \approx S \left(X(t) - \frac{\beta}{\alpha} \right) \quad (33)$$

for some ‘ S ’.

Taking the Laplace Transform of the latter two expressions and solving for that of X ,

$$\mathcal{L}(X(t), t, s) = \frac{\beta(S + \alpha)}{\alpha s (\alpha \mathcal{L}(\rho(t), t, s) + S)} \quad (34)$$

In Eq.(33), the Laplace Transform of $\rho(t)$, with reversed sign in t , is the moment generating function of that distribution, which satisfies Jensen’s inequality

$$M_X(t) \geq \exp[\mu t] \quad (35)$$

where μ is the first moment of ρ .

Based on this, in Eq.(33) we make the usual crude ‘physics’ approximation $\mathcal{L}(\rho(t), t, s) \approx \exp[-\langle \tau \rangle s]$, take the total inverse Laplace Transform, and solve the resulting relation for S , finding

$$X(t) = \frac{\beta(1 - e^{St})}{\alpha} = \frac{\left(1 - \left(-\frac{s}{\alpha}\right)^{-1 - \lfloor \frac{t}{\tau} \rfloor}\right)\beta}{\alpha} \\ S = \frac{\left(1 + \lfloor \frac{t}{\tau} \rfloor\right) W\left(m, -\frac{\alpha t}{1 + \lfloor \frac{t}{\tau} \rfloor}\right)}{t} \quad (36)$$

where, for simplicity, we have omitted the brackets around τ .

The limit $t \rightarrow \infty$ recovers the second expression of Eq.(19) in terms of the average $\langle \tau \rangle$

$$S \rightarrow \frac{W(m, -\alpha \langle \tau \rangle)}{\langle \tau \rangle} \quad (37)$$

real-valued only for $m = 0, -1$ and $\alpha \langle \tau \rangle \leq \exp[-1]$.

There is, however, less to this argument than first appears, since $\exp[-\mu s]$ from the Jensen inequality is, in fact, the Laplace Transform of the Dirac delta function $\delta(t - \mu)$. Nonetheless, the development is instructive.

Numerical solutions for general probability distributions can sometimes be explored using a fast computer algebra program, giving critical values of $\alpha \langle \tau \rangle$. This is done by solving the first two parts of Eq.(33) using the Laplace Transform method to determine an expression for $X(t)$, then evaluating S from the complicated expressions that are the numerical forms of the appropriate Laplace and inverse Laplace transforms, finding

$$S = \frac{dX/dt}{(X(t) - \beta/\alpha)} \quad (38)$$

and taking appropriate limits.

The method is not without drawbacks.

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